GENARCHAI: AN AI-POWERED PLATFORM FOR GENERATIVE ARCHITECTURAL DESIGN

A Project Report

Submitted by

FATHIMMA SANA (*MES22AD024*)

to

The APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

in

Artificial Intelligence and Data Science

Department of Artificial Intelligence and Data Science

MES College of Engineering Kuttippuram

Thrikkanapuram South P.O, Malappuram Dt, Kerala, India 679582

2025 - 2026

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MES COLLEGE OF ENGINEERING KUTTIPPURAM

CERTIFICATE

This is to certify that the project report entitled **" GenArchAI: an AI-Powered Platform for Generative Architectural Design"**, submitted by **Fathimma Sana**, to the APJ Abdul Kalam Technological University, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Artificial Intelligence and Data Science, is a bonafide record of the project work carried out under our guidance and supervision. This report, in any form, has not been submitted to any other University or Institute for any purpose .

Project Guide:	**Project Coordinator:**	**Head of Department:**
Mr. Sherikh K K	Ms. Sherin Shana	Dr. Govindaraj. E
Assistant Professor	Assistant Professor	Associate Professor
Dept of ADS	Dept of ADS	Dept of ADS

DECLARATION

We hereby declare that the project report named **" GenArchAI : An AI - Powered Platform for Generative Architectural Design"** submitted to partial fulfillment of the requirements for the award of the Bachelor of Technology degree of APJ Abdul Kalam Technological University, Kerala, is a true work carried out under the supervision of **Mr. Sherikh K K**, Assistant Professor, Department of Artificial Intelligence and Data Science. This submission represents our ideas in our own words and where ideas or words of others have been included, We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Kuttippuram FATHIMMA SANA

Date: 21/04/2026

ACKNOWLEDGEMENT

We are grateful to *Dr. Rahumathunza I, Principal, MES College of Engineering, Kuttippuram*, for providing the right ambiance to complete this project. We would also like to extend our sincere gratitude to *Dr. Govindaraj.E, Head of Department, Artificial Intelligence and Data Science, MES College of Engineering, Kuttippuram.*

We are deeply indebted to the project coordinator, *Ms. Sherin Shana, Assistant Professor, Department of Artificial Intelligence and Data Science* for her continued support.

It is with great pleasure that we express our deep sense of gratitude to our project guide, *Mr. Sherikh K K, Assistant Professor, Department of Artificial Intelligence and Data Science*, for his guidance, supervision, encouragement and valuable advice in each and every phase.

We would also like to express our sincere thanks and gratitude to all staff members of the department, our friends and family members for their cooperation, positive criticism, consistent support and consideration during the preparation of this work.

FATHIMMA SANA

ABSTRACT

GenArchAI is an AI-powered web platform developed to simplify architectural design, renovation, and cost estimation using modern Artificial Intelligence techniques. Traditional architectural planning requires professional expertise, expensive software, and long development time, which makes it difficult for common users to create quality building designs. GenArchAI provides a user-friendly system where users can generate building designs from text prompts or sketches, view structures from multiple angles, estimate construction cost in Indian Rupees, renovate existing rooms, and receive architectural guidance through an AI chatbot. The system is implemented as a full-stack web application using React.js for the frontend and Node.js with Express.js for the backend, integrating multiple AI services for image generation, vision analysis, and conversational assistance. By combining Generative AI, computer vision, and cloud-based storage, the platform provides fast, affordable, and interactive architectural visualization, making professional design tools accessible to homeowners, students, and developers.

Keywords: Generative AI, Architectural Design, Image Generation, Cost Estimation, Web Application, Artificial Intelligence, Design Automation.

Contents

List of Figures

Abbreviations

GenArchAl	Generative Architectural Artificial Intelligence
GenAI	Generative Artificial Intelligence
ANN	Artificial Neural Network
BIM	Building Information Modeling
CAD	Computer-Aided Design
AI	Artificial Intelligence
CV	Computer Vision

Chapter 1

Introduction

GenArchAI is an advanced AI-powered platform designed to revolutionize the field of architectural design, planning, and renovation. Architecture is an industry that has traditionally relied on manual processes, from initial concept sketches to detailed blueprints, from labor-intensive cost estimation spreadsheets to time-consuming client presentations. Conventional architectural workflows require significant expertise, expensive software tools such as AutoCAD, Revit and SketchUp, and long development cycles that may take weeks or even months. For individual homeowners, especially in developing regions, the cost of hiring a professional architect along with the lengthy design process creates a major barrier to obtaining high-quality architectural planning.

GenArchAI addresses this limitation by providing an AI-powered architectural design generation platform that makes architectural intelligence accessible to everyone. The system enables homeowners, architecture students, designers, and small-scale developers to generate professional-quality building designs, renovation layouts, cost estimates, and architectural guidance through a modern web-based interface powered by Generative Artificial Intelligence.

From a user perspective, GenArchAI works like a personal architect available at any time. A user can describe the type of building they want, such as a modern two-floor house with multiple rooms, and the system generates realistic architectural images based on the description. Users can also upload hand-drawn sketches, and the platform converts them into professional architectural renderings. The system can generate different views of the structure, including front, side, rear, and aerial views, helping users visualize the design before construction. In addition, the platform provides estimated construction costs, suggestions for budget optimization, and an AI chatbot that answers architecture-related questions, making the entire planning process simple and interactive.

Technically, GenArchAI is implemented as a full-stack web application built using a React.js frontend and a Node.js/Express.js backend. The system integrates multiple Artificial Intelligence services through a unified API layer. Uploaded sketches are analyzed using a Vision Analysis Pipeline that extracts structural information from the image. The extracted description is then passed to an Image Generation Pipeline, where generative models produce realistic architectural renderings. A Conversational AI module is used to provide real-time assistance and answer user queries related to design, renovation, and construction.

The platform also includes a Cost Estimation Engine that calculates approximate construction expenses based on area, materials, and design type, helping users plan their projects within budget. Data storage is handled using cloud-based databases, and generated images are stored securely using cloud storage services. Real-time interaction between frontend and backend is enabled through modern web communication technologies, ensuring a smooth and responsive user experience.

The motivation behind developing GenArchAI comes from the growing demand for affordable and efficient architectural planning solutions. The construction industry is one of the largest sectors, yet many residential projects are completed without proper design guidance due to cost and accessibility issues. This often results in poor space utilization, increased expenses, and designs that do not meet modern standards. GenArchAI aims to bridge this gap by providing an intelligent, automated, and user-friendly system that allows anyone to create high-quality architectural designs without requiring professional-level expertise.

In addition to practical applications, the platform also has educational and professional value. Students can use it to visualize architectural concepts, designers can use it to generate quick proposals, and homeowners can explore multiple design options before making decisions. By combining Artificial Intelligence, Machine Learning, and Computer Vision, GenArchAI represents a step toward the future of smart architectural design systems.

Chapter 2

Review of Literature

The Architecture, Engineering, and Construction (AEC) industry is undergoing a paradigm shift driven by the rapid evolution of **Generative Artificial Intelligence (GenAI)**. These tools are transforming the design process, offering solutions that range from creative ideation to technical documentation and project management. This literature review synthesizes the findings of five recent academic papers, exploring the core applications of GenAI in modern architecture. The reviewed works collectively demonstrate a shift towards intelligent, data-integrated design workflows, focusing on translating abstract intent, integrating with Building Information Modeling (BIM), ensuring project feasibility, and creating intuitive human-AI interfaces.

Generative AI-powered architectural exterior conceptual design based on the design intent [1]. Shi et al. explored how to ensure GenAI outputs remain relevant and controllable by the architect. Their research proposes a sophisticated framework where abstract, qualitative design intent (e.g., specific style, material selection, or overall mood) is successfully translated into quantifiable parameters. These parameters then govern the GenAI model, directing it to produce architectural exterior concepts that are not merely novel, but are also highly aligned with the designer's vision and project brief. This paper is a crucial contribution to the field, as it moves GenAI beyond simple image generation and establishes a foundation for using it as a controlled design partner that adheres to specific aesthetic and functional constraints.

Generative AI-powered parametric modeling and BIM for architectural design and visualization [2]. Ko et al. addressed the critical gap between AI-generated visuals and their practical use in construction workflows. They investigated the integration of GenAI with parametric modeling and BIM. Their methodology focuses on ensuring that the models generated by the AI are inherently structured, data-rich, and immediately compatible with BIM software. By producing models that contain embedded parameters and building information, the AI output transitions from a static conceptual image to a dynamic, usable construct. This work is essential for realizing the full potential of AI in the AEC sector, as it guarantees a smooth, automated workflow from the initial design concept straight through to technical documentation and project execution, thereby improving data consistency and reducing manual rework.

Prediction of cost and duration of building construction using artificial neural network [3]. Ujong et al. utilized an Artificial Neural Network (ANN) to analyze historical project data and provide accurate predictions for the cost and duration of building construction. While this study predates the widespread use of GenAI for pure design, its findings are foundational for integrating financial and temporal feasibility checks into a generative platform. The research validates the capability of AI models to provide reliable forecasts based on complex project variables. For a comprehensive AI design system, this predictive capacity is vital: it serves as the core of a 'Smart Budget' feature, allowing architects to receive real-time cost feedback on their AI-generated designs and enabling the optimization of form and material selection to fit prescribed fiscal and scheduling constraints.

Sketch-to-Architecture: Generative AI-aided Architectural Design [4]. Li et al. explored the most intuitive application of GenAI in the ideation phase: the "Sketch-to-Architecture" paradigm. This research demonstrates how a designer's rudimentary freehand sketch can serve as the primary, low-barrier-to-entry input for the generative model. The AI then rapidly translates this rough input into detailed architectural outputs, including high-fidelity 3D visualizations, elevation views, and fully-formed floor plans. The significance of this work lies in its ability to drastically reduce the time and effort required to move from an abstract idea to a concrete visual concept, fundamentally accelerating the creative loop. By leveraging the designer's most natural input, this approach makes high-quality design visualization accessible and fluid.

AI Chatbots in Architectural Design: Exploring the Potential and Limitations in Conceptual Image Development [5]. Yim et al. investigated the pivotal role of ChoiceAI Chatbots as a natural-language interface for complex generative design tools. This study focuses on how conversational AI can be used to guide users through the process of conceptual image development. The chatbot is framed as a design assistant that can interpret complex, ambiguous user prompts, provide context-aware suggestions, and offer feedback regarding design constraints, styles, and budget considerations. This research highlights the importance of enhancing the human-computer interaction (HCI) aspect of GenAI. By offering an accessible and conversational method to control powerful generative algorithms, this work makes sophisticated tools easier to adopt and integrate into the architect's daily workflow, facilitating iterative design refinement through dialog.

The Reviewed Literature confirms GenAI's profound impact on architectural practice, establishing it as an indispensable co-creator across the project lifecycle. This transformation is driven by several innovations:

1. The ability to translate abstract design intent into quantifiable rules for generative models, ensuring creative alignment.
2. The production of technically robust, BIM-ready outputs for seamless construction integration.
3. The application of predictive modeling to ensure designs meet real-world cost and duration feasibility
4. GenAI accelerates initial design by converting rough sketches into detailed 3D models.
5. The maintenance of accessibility through conversational AI Chatbots that provide project guidance.

These distinct functions are collectively driving the shift toward unified, efficient, and intelligent design platforms.

Chapter 3

System Analysis

3.1 Existing System

Current architectural design tools fall into three main categories, each with several limitations. Although these tools assist in architectural visualization and drafting, they often require significant manual effort and technical expertise.

3.1.1 Traditional CAD Software

Tools such as AutoCAD, Revit, and SketchUp are widely used by professional architects and designers to create detailed 2D drawings and 3D models. These tools provide precise control over architectural components and support complex building designs. However, they require extensive training and experience to operate efficiently. The design process is mostly manual, making it time-consuming for architects to generate multiple design alternatives. In addition, these tools involve expensive licensing costs and usually do not include AI-based design generation or automated cost estimation for Indian construction markets.

3.1.2 Online Architecture Platforms

Platforms like Planner5D and HomeByMe allow users to create basic floor plans and interior layouts through drag-and-drop interfaces. These tools are easier to use compared to professional CAD software and are suitable for beginners or homeowners. However, they are mainly based on predefined templates and limited design libraries. As a result, Customization options are restricted and users cannot easily generate unique building designs. Furthermore, these systems lack AI-powered features such as sketch interpretation, intelligent design suggestions, and accurate construction cost estimation.

3.1.3 AI Image Generators

General AI tools such as Midjourney, DALL-E, and Stable Diffusion are capable of generating realistic images from text prompts. While they can produce visually appealing architectural images, they are not specifically designed for architectural workflows. These tools do not understand building measurements, structural constraints, or architectural standards. They also cannot analyze hand-drawn sketches, generate consistent multi-view building designs, estimate construction costs, or provide project management features required for real architectural planning.

Overall, the existing systems either focus on manual design tools or generic AI image generation. None of these solutions provide a unified platform that combines AI-based design generation, sketch analysis, cost estimation, and intelligent architectural guidance in a single system.

3.1.4 Limitations of Existing Systems

- High cost and steep learning curve for professional CAD tools.
- Lack of cost estimation based on Indian construction rates.
- No integration between design generation, cost estimation & architectural guidance.
- No single platform combining AI design, chatbot assistance & project management.

3.2 Proposed System

GenArch AI is an AI-powered architectural design platform that integrates multiple intelligent services into a single web application.

The proposed system provides the following features:

- **AI Design Generation:** Generates architectural designs from text prompts or sketches.
- **Multi-View Rendering:** Creates multiple views of a building such as front, side, rear, and aerial views.

- **Sketch-to-Design Conversion:** Converts hand-drawn sketches into realistic building designs using AI.
- **AI Chatbot:** Provides architectural guidance and cost advice based on Indian construction practices.
- **Cost Estimation:** Estimates construction cost using CPWD-based rates.
- **Room Renovation Module:** Allows users to upload a room image and generate renovated designs.
- **Budget Optimizer:** Suggests cost-saving alternatives for construction.
- **Design Comparison:** Enables comparison between multiple design options.
- **PDF Report Generation:** Generates downloadable project brochures.
- **Project Management:** Allows users to create, store, and manage architectural projects.

3.3 Objectives of Proposed System

1. To develop an AI-based system that generates architectural designs from text descriptions and sketches.
2. To implement multi-view building visualization.
3. To provide construction cost estimation based on Indian market rates.
4. To integrate conversational AI for architectural guidance.
5. To enable renovation design suggestions using AI.
6. To provide budget optimization recommendations.
7. To support project management and design comparison.
8. To create a user-friendly web interface accessible to both professionals and non-experts.

3.4 Advantages of Proposed System

- AI-based design generation reduces manual work and design time.
- Cost estimation is based on Indian construction standards.
- Sketch analysis allows quick conversion from concept to design.
- Multiple users can collaborate on the same project.
- No expensive architectural software licenses are required.
- Generates professional reports for client presentations.

3.5 Requirement Specification

3.5.1 Functional Requirements

- Users can generate building designs from text prompts.
- Users can upload sketches to generate architectural designs.
- The system generates multiple building views.
- The system estimates construction cost.
- AI chatbot provides architectural guidance.
- Users can upload room images for renovation suggestions.
- Users can export designs as PDF reports.
- Users can create and manage architectural projects.

3.5.2 Non-Functional Requirements

- The system should generate designs within a short response time.
- The interface must be simple and user-friendly.
- The system should support multiple users simultaneously.
- API keys and user data must be stored securely.
- The application should work on both desktop and mobile browsers.

3.6 Feasibility Study

3.6.1 Technical Feasibility

The proposed system uses widely available technologies such as React.js for the frontend and Node.js with Express for the backend. AI services including image generation models, Gemini Vision, and conversational AI are integrated through APIs. Firebase Firestore is used for data storage, and Socket.IO enables real-time collaboration. These technologies make the system technically feasible.

3.6.2 Economic Feasibility

The project uses mostly open-source tools and free-tier AI services. Development can be performed using existing hardware and free software libraries. Therefore, the overall development cost is minimal, making the project economically feasible.

3.6.3 Operational Feasibility

The system provides a simple and interactive interface that allows users to generate architectural designs without prior technical knowledge. The platform can be deployed easily on standard cloud hosting services, and maintenance is simplified due to the modular system design.

3.6.4 Legal and Regulatory Feasibility

AI-generated images are produced using models that allow academic use. The system does not collect sensitive personal information, ensuring privacy protection. A disclaimer is included stating that generated designs should be reviewed by qualified architects before real construction.

Chapter 4

Design and Development

4.1 System Design

The system design of **GenArchAI** is structured to support AI-powered architectural design generation, renovation visualization, cost estimation, and conversational assistance. The design follows a modular and scalable approach using a three-tier client–server architecture with external AI service integration. The system is developed as a full-stack web application to ensure real-time interaction, high performance, and easy scalability.

4.1.1 Architectural Design

GenArchAI follows a three-tier architecture consisting of Presentation Tier, Application Tier, and Data/External Services Tier, as illustrated in Figure 4.1. This layered design approach aligns with modern architectural frameworks used in AI-integrated systems [7].

- **Presentation Tier (Frontend):** Developed using React.js with Vite and Framer Motion. Provides interfaces such as Landing Page, Dashboard, Design View, Project Workspace, Renovation Studio, Budget Optimizer, Presentation Mode, and AI Chat. Axios is used for API communication through REST and Socket.IO.

- **Application Tier (Backend):** Implemented using Node.js and Express.js with middleware for CORS, JSON parsing, rate limiting, and static files. Route modules handle design, renovation, cost, chat, and project operations. Services integrate HuggingFace, Gemini, Groq, Cloudinary, and Cost Engine. Utilities like Sharp and DataNormalizer handle image processing and formatting.

- **Data / External Services Tier:** Firebase Firestore stores project data and Cloudinary stores images. HuggingFace FLUX.1 generates designs, Gemini analyzes sketches, and Groq LLaMA 3.1 powers the chatbot. An in-memory fallback store is used when external services are unavailable.

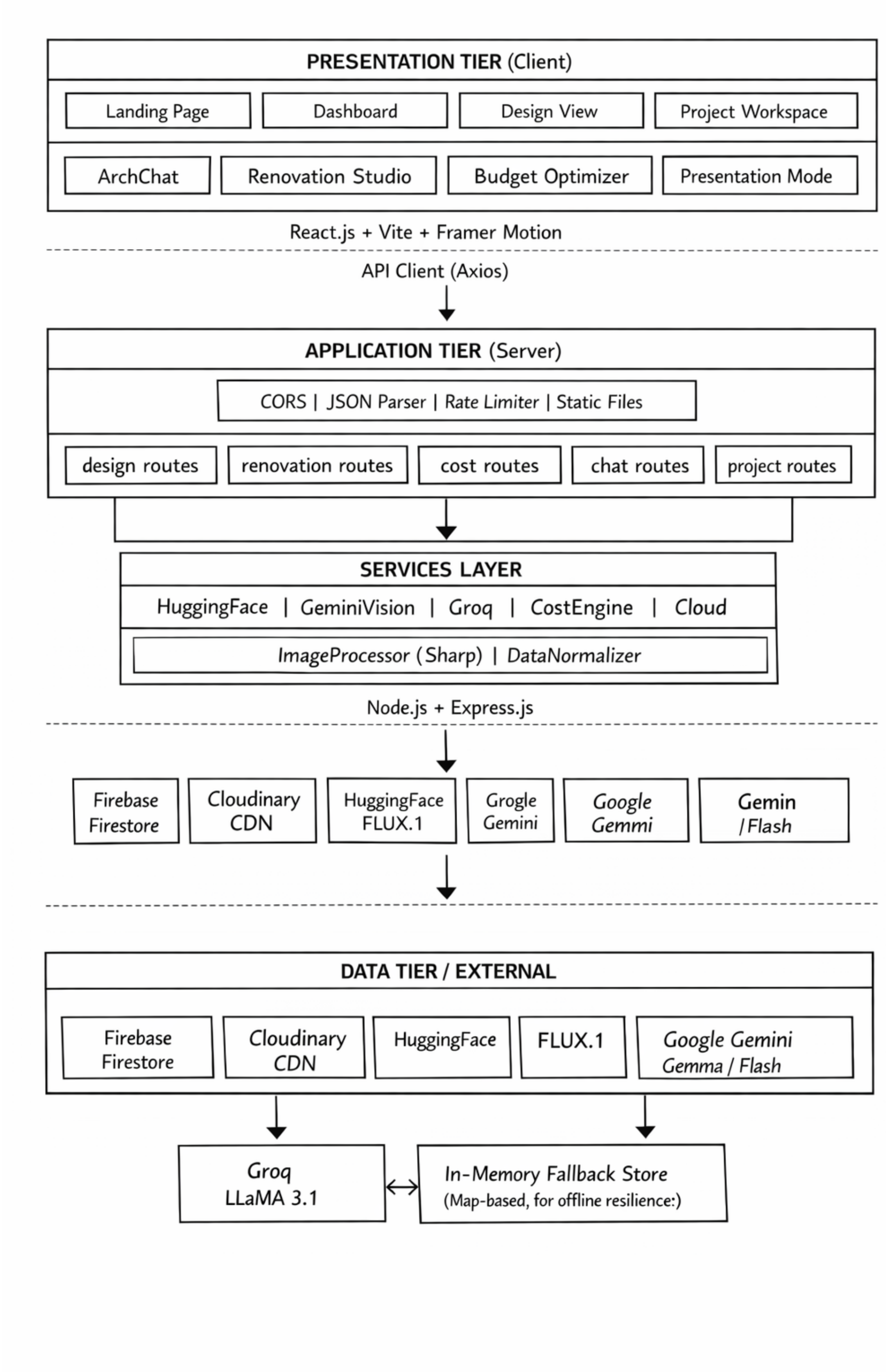

Figure 4.1: System Architecture Flow Diagram

4.2 System Modules

The GenArchAI platform is divided into multiple modules that work together to generate designs, perform renovation visualization, estimate cost, and provide AI assistance.

4.2.1 AI Design Generation Module

This is the core module of the system. Users enter a text prompt or upload a sketch. The frontend sends a request to the backend API. If a sketch is uploaded, the Gemini Vision service analyzes the structure. The structural prompt and style modifiers are sent to the HuggingFace FLUX.1 model. The generated image is compressed using Sharp and uploaded to Cloudinary. CostEngine calculates cost and scores. The result is stored in Firebase Firestore.

Pipeline Steps

1. User enters prompt or uploads sketch
2. Request sent to /api/generate-design
3. Middleware validates request
4. Gemini Vision analyzes sketch
5. HuggingFace generates image
6. Sharp compresses image
7. Cloudinary stores image
8. CostEngine calculates cost
9. Firestore saves design
10. Socket.IO sends status updates
11. Response returned to frontend

4.2.2 Room Renovation Module

Users upload a photo of a room and select a renovation style. Gemini Vision extracts layout information. HuggingFace generates a new image with the same layout but different style. The result is stored in Cloudinary and Firestore. This module allows users to visualize renovation before construction.

4.2.3 Cost Estimation Module

The Cost Engine calculates approximate construction cost in Indian Rupees.

The estimation depends on:

- Area in square feet
- Number of floors
- Quality level
- Location type
- Style type

The system generates a detailed breakdown including foundation, structure, electrical, plumbing, flooring, painting, kitchen, bathroom, exterior, landscaping, and permits. Budget optimizer suggests ways to reduce cost.

4.2.4 AI Chatbot Module

The chatbot allows users to ask architecture-related questions.

It uses Groq LLaMA 3.1 model.

Features:

- Context-aware responses
- Indian cost format
- Project-aware answers
- History memory
- Firestore chat storage

The chatbot helps users with design decisions, materials, and budget planning.

4.2.5 Budget Optimizer Module

If the cost exceeds budget, the system suggests optimizations.

Examples:

- Reduce floors
- Change style
- Change location
- Reduce area

- Use prefabricated materials

Savings are calculated automatically.

4.2.6 Design Comparison Module

Users can compare two designs.

Comparison includes:

- Cost
- Space efficiency
- Sustainability
- Luxury score

This helps users choose the best design [8].

4.3 Development Process

4.3.1 AI Model Integration

GenArchAI integrates three main AI services [6].

- **Image Generation** HuggingFace FLUX.1 model is used for architecture rendering.
- **Vision Analysis** Google Gemini analyzes sketches and rooms.
- **Conversational AI** Groq LLaMA 3.1 is used for chatbot.

4.3.2 Backend Development

Backend is built using Node.js and Express.js.

Layers:

- Entry point (index.js)
- Routes layer
- Services layer
- Utils layer

Middleware used:

- cors
- express.json

- multer
- rate limiter

4.3.3 Frontend Development

Frontend is built using React.js.

Features:

- React Router navigation
- Axios API client
- Framer Motion animations
- Dark theme UI
- Component-based design

4.4 Database Design

Data is stored in Firebase Firestore.

Entities:

- Users
- Projects
- Designs
- Costs
- Chat history

Chapter 5

Coding

5.1 Front End Development

The front end of the system is developed using React.js with the Vite build tool, which enables faster development and efficient performance. React provides a component-based architecture that allows the interface to be divided into reusable modules such as navigation bars, dashboards, and design panels. This approach improves maintainability and scalability of the application.

The user interface is designed to provide an interactive experience where users can easily generate architectural designs, manage projects, and visualize generated outputs. React Router is used to implement page navigation within the application without requiring full page reloads, which improves responsiveness and overall user experience. The interface also supports dynamic updates based on user input and API responses.

Sample React Component

```
<Routes>
  <Route path="/" element={<LandingPage />} />
  <Route path="/dashboard" element={<Dashboard />} />
</Routes>
```

5.2 Backend Development

The backend of the system is implemented using Node.js with the Express.js framework. The backend server is responsible for handling client requests, processing user inputs, communicating with AI services, and returning generated results to the frontend. Express provides a lightweight and flexible framework for creating RESTful APIs and managing server-side operations.

The backend also manages routing, request validation, and data processing. It ensures that user inputs such as prompts, images, and design parameters are properly handled before sending them to AI models. Additionally, middleware functions are used to process JSON data and control request flow within the server.

Basic Server Setup

```
const app = require("express")();
app.use(express.json());
app.listen(5000);
```

5.3 API Integration

The system integrates several Artificial Intelligence APIs to provide advanced functionality. These APIs enable the platform to perform tasks such as image generation, sketch analysis, and conversational assistance. Hugging Face models are used to generate architectural images based on text prompts, while Google Gemini Vision is used to analyze sketches and extract structural details.

Additionally, the Groq LLaMA model is used to power the AI chatbot, which provides architectural guidance and design suggestions to users. By integrating these APIs, the system combines multiple AI capabilities into a single platform, enabling users to generate designs and obtain insights in real time.

Example API Call

```
const response = await axios.post("/api/generate-design", {
  prompt: userPrompt,
  style: selectedStyle
});
```

5.4 Image Generation Workflow

The image generation process begins when a user enters a design prompt or uploads a sketch through the user interface. The backend receives this input and constructs a suitable prompt for the AI image generation model. If a sketch is provided, it is analyzed to understand the structure of the building.

The processed prompt is sent to the image generation model, which produces a photorealistic architectural image. The generated image is optimized using image processing techniques and stored in cloud storage. Finally, the image URL and related design information are returned to the frontend and displayed to the user.

Image Processing Example

```
const image = await generateImage(prompt);
const compressed = await sharp(image)
    .resize(1024,1024)
    .jpeg({quality:85});
```

5.5 Security and Validation

Security and validation mechanisms are implemented to ensure reliable system operation. API keys used for external AI services are stored securely using environment variables so they are not exposed in client-side code.

The system also applies rate limiting to AI endpoints to prevent excessive requests. Input validation ensures that user data is processed safely and helps maintain system stability.

Rate Limiting Example

```
const limiter = rateLimit({
  windowMs: 60000,
  max: 10
});
app.use("/api", limiter);
```

Chapter 6

Software Requirements Specification

6.1 Scope

GenArchAI is a web-based AI-powered architectural design generation system that allows users to generate building designs, estimate construction costs, renovate rooms, and obtain architectural guidance using Artificial Intelligence.

The system provides the following features:

- Generate photorealistic building designs from text or sketches
- View designs from multiple camera angles
- Estimate construction cost in Indian Rupees
- AI chatbot for architectural guidance
- Room renovation using AI style transfer
- Budget optimization suggestions
- Design comparison
- PDF brochure export
- Real-time collaboration

The system is intended for homeowners, students, designers, and small developers.

6.2 Specific Requirements

6.2.1 User Interface Requirements

1. Landing Page with project introduction and start button

2. Dashboard showing project list with create, edit, delete options
3. Design View with image display, cost chart, and score radar
4. Project Workspace with tabs for design, renovation, chat, and budget
5. Architect Chat with message bubbles and typing indicator
6. Renovation Studio with image upload and style selector
7. Budget Optimizer with cost slider and suggestions
8. Presentation Mode for slideshow display
9. Upload Design form with parameters and sketch upload
10. Responsive design for mobile and desktop

6.2.2 Backend Requirements

1. REST API under /api
2. File upload support (max 10MB)
3. AI services handled using service classes
4. Firestore with fallback memory storage
5. Image compression using Sharp
6. Socket.IO for real-time updates
7. Rate limiting for AI requests
8. Error handling to prevent crashes

6.2.3 Workflow

1. User opens landing page
2. Creates project in dashboard
3. Enters workspace
4. Generates design
5. System runs AI pipeline
6. Image displayed
7. Cost calculated
8. User checks scores
9. User optimizes budget

10. User exports PDF
11. User chats with AI

6.3 Technology Stack

The system is built using a modern full-stack architecture that integrates frontend, backend, cloud services, and multiple AI models. Each component is carefully selected to ensure performance, scalability, and seamless user experience. The detailed technologies used in each layer of the system are shown in Table 6.1.

Layer	Technology	Purpose
Frontend	React.js	UI
Build Tool	Vite	Development server
Routing	React Router	Navigation
Animation	Framer Motion	UI animation
Charts	Recharts	Cost charts
HTTP Client	Axios	API calls
Backend	Node.js	Server runtime
Framework	Express.js	API routing
Database	Firebase Firestore	Storage
Image Processing	Sharp	Compression
Upload	Multer	File upload
PDF	PDFKit	Brochure export
Realtime	Socket.IO	Live updates
Cloud	Cloudinary	Image storage
AI Image	HuggingFace	FLUX.1 model
AI Vision	Google Gemini	Sketch analysis
AI Chat	Groq LLaMA	Chatbot
UUID	uuid	IDs

Table 6.1: Technology Stack

6.4 Setup and Usage

6.4.1 Server Setup

```
cd server
npm install
npm run dev
```

6.4.2 Client Setup

```
cd client
npm install
npm run dev
```

6.5 Conclusion

GenArchAI is a modular AI-powered architecture design system that integrates multiple AI services into a single platform. The system provides design generation, renovation, cost estimation, chatbot guidance, and visualization tools. The architecture ensures scalability, reliability, and ease of use.

Chapter 7

Implementation

7.1 User Interface Implementation

The user interface is implemented using React.js with component-scoped CSS following a dark themed glassmorphism design, as shown in Figure 7.1.

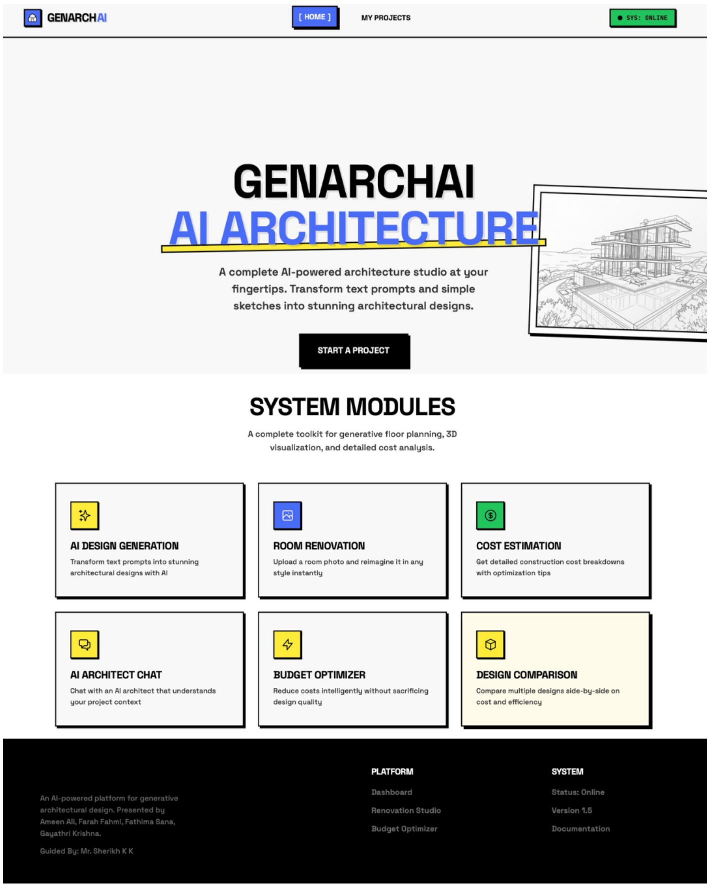

Figure 7.1: GenArchAI User Interface – Landing Page and System Modules

7.1.1 Main Components

- LoadingScreen – animated splash screen
- Navbar – responsive navigation bar
- GlassCard – reusable UI container
- CostChart – pie chart for cost display
- ScoreRadar – radar chart for design scores
- AICritique – AI feedback panel
- LoadingOverlay – full screen loading indicator

7.2 GenArchAI Design Generation Implementation

The Design View page is responsible for design generation and visualization, as illustrated in Figure 7.2. Key functionalities include multi-view navigation, a comprehensive design input form, real-time design generation powered by Socket.IO, cost charts with score radar visualization, PDF brochure export, image download, and the ability to mark designs as favorites.

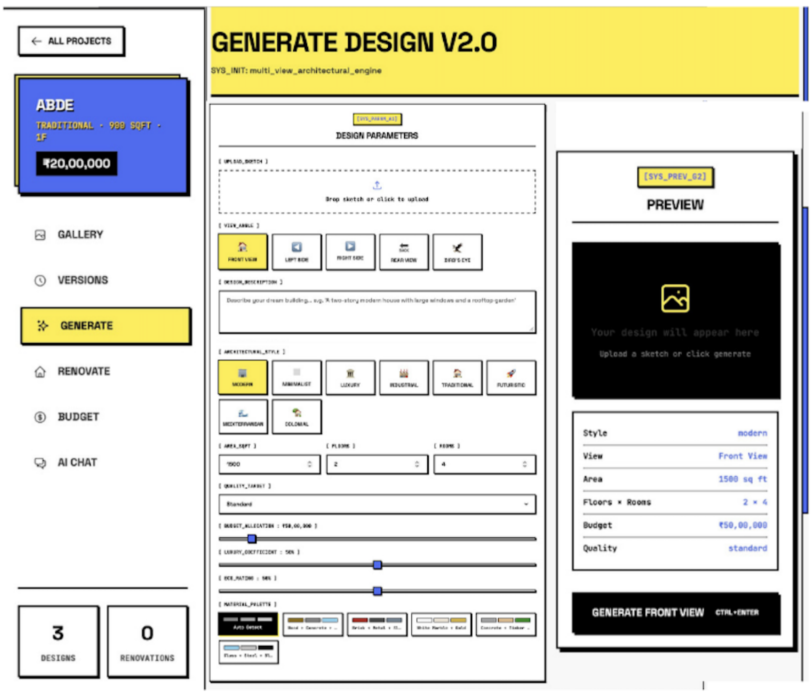

Figure 7.2: AI Design Generation Page

7.3 Renovation System Implementation

The renovation module allows users to upload a room image and generate a new style, as shown in Figure 7.3.

Features:

- Drag and drop upload
- Style selection
- Additional prompt input
- Before / after comparison
- Renovation history

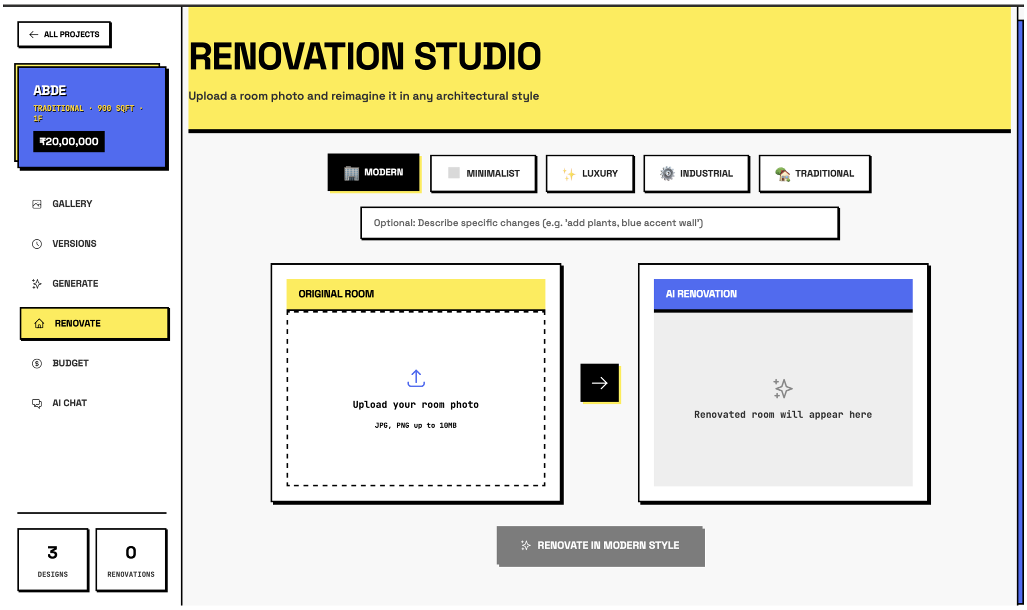

Figure 7.3: GenArchAI Renovation Studio Interface

7.4 Chatbot Implementation

The chatbot allows users to ask architecture questions, as shown in Figure 7.4.

Features:

- Chat message list
- Project context display
- Message input box
- Typing indicator

- Chat history storage
- Clear history option

Figure 7.4: GenArchAI AI Architect Chat Interface

7.5 Cost Estimation Implementation

Cost estimation is computed on the server using CostEngine, which processes design inputs and calculates accurate cost breakdowns based on predefined rules and parameters, as shown in Figure 7.5.

Displayed using:

- CostChart – a pie chart that visually represents the distribution of different cost components for better understanding.
- BudgetOptimizer – an interactive cost slider that allows users to adjust budget limits and view optimized suggestions accordingly.
- ScoreRadar – a radar chart that illustrates overall design performance across multiple evaluation metrics.
- INR formatting utilities – functions used to display all cost values in a properly formatted Indian Rupee (INR) style for clarity.

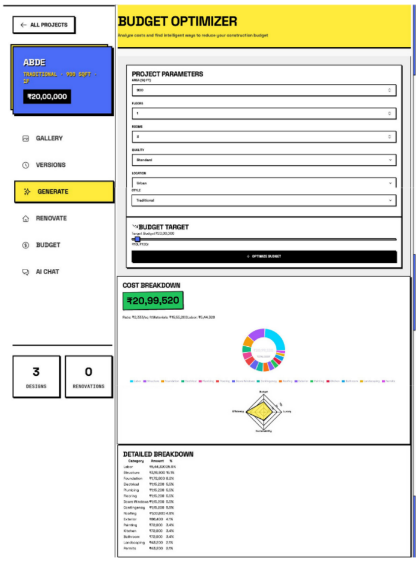

Figure 7.5: GenArchAI Budget Optimizer Interface

7.6 API and Database Integration

7.6.1 Firestore Collections

The system uses Firebase Firestore as the cloud database for storing project information, generated designs, and chatbot conversations. Firestore stores data in collections and documents, which allows flexible and scalable data management.

- projects – project data
- designs – generated images
- chat_history – chat messages
- users – user session or device information (if available)
- settings – application configuration and default parameters

Firestore is used because it supports real-time updates, easy integration with Node.js, and secure cloud storage for web applications.

7.6.2 API Endpoints

The system provides API endpoints for design generation, project management, and AI features, as shown in Table 7.1.

Method	Endpoint	Description
POST	/api/generate-design	Generate design
POST	/api/generate-view	Generate view
POST	/api/generate-style	Apply style
GET	/api/design/:id	Get design
POST	/api/designs/compare	Compare designs
POST	/api/renovate	Renovate room
POST	/api/estimate-cost	Estimate cost
POST	/api/optimize-budget	Optimize budget
POST	/api/chat	Chat message
GET	/api/projects	List projects
POST	/api/projects	Create project
PUT	/api/projects/:id	Update project
DELETE	/api/projects/:id	Delete project
GET	/api/health	Health check

Table 7.1: API Endpoints

Chapter 8

Result and Discussion

8.1 System Output

The GenArch AI system generates several outputs that assist users in architectural design and planning. The system integrates AI models, image generation techniques, and cost estimation modules to support architectural design.

Photorealistic Architectural Renderings: The system generates high-quality architectural images based on user-defined prompts and design preferences. Each rendering is produced at a resolution of 1024×1024 pixels, ensuring clarity and visual appeal. These realistic images help users better visualize the final structure before construction begins, as shown in Fig. 8.1.

Figure 8.1: Photorealistic architectural rendering generated by the system

Multi-View Building Visualizations: The system provides multiple perspectives of the same building design to enhance understanding. These views include front, left, right, rear, and aerial angles. Structural prompts are reused to maintain consistency across all generated views. This feature allows users to analyze the design from different orientations and make informed decisions as shown in Fig. 8.2.

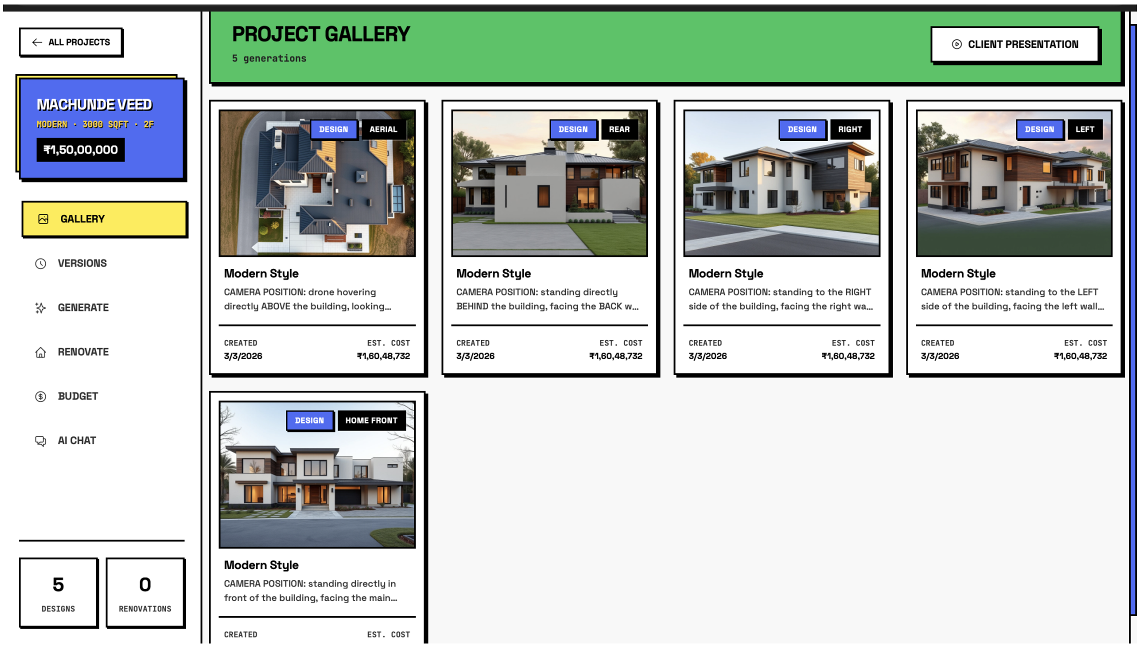

Figure 8.2: Multi-view visualization of the building design

Detailed Cost Estimates: The application generates approximate construction cost estimates in Indian Rupees. It provides a detailed breakdown of expenses including foundation, structure, roofing, and finishing works. Additional components such as electrical, plumbing, flooring, and interior elements are also included. This helps users plan their budget effectively and understand cost distribution.

Budget Optimization Suggestions: The system offers intelligent suggestions to reduce overall construction costs. It recommends alternative materials and design modifications that maintain quality while lowering expenses. These suggestions are generated based on the initial design and estimated budget. This feature helps users make cost-effective decisions without compromising functionality or aesthetics, as shown in Fig. 8.3.

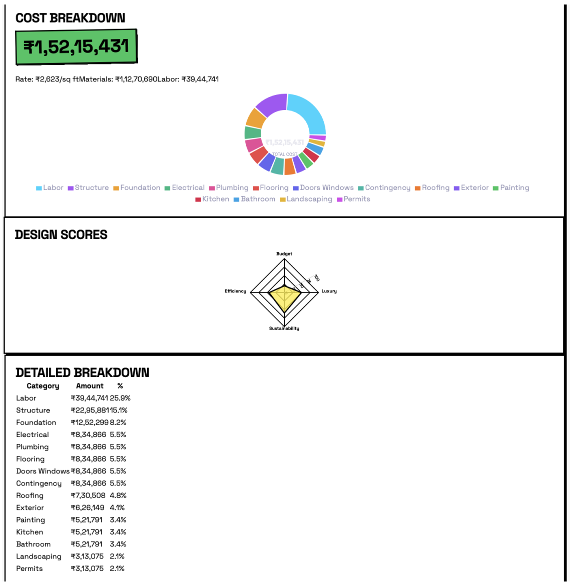

Figure 8.3: Detailed cost estimation along with budget optimization suggestions

AI Chatbot Assistance: An AI chatbot provides architectural guidance during the planning and design process. It answers user queries related to layouts, materials, and design choices in real time. This interactive support helps users make informed decisions efficiently, as shown in Fig. 8.4.

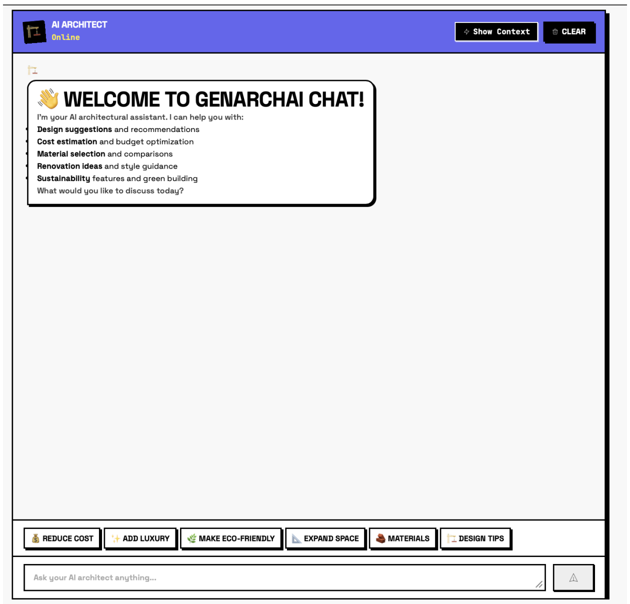

Figure 8.4: AI chatbot assisting in architectural planning

PDF Report Generation: The system generates PDF brochures containing design images, project details, and cost summaries that can be shared with clientss, as shown in Fig. 8.5.

Figure 8.5: Generated PDF report sample

8.2 System Testing and Evaluation

To evaluate the performance and usability of the GenArch AI system, a structured testing process was conducted involving multiple users. Since the system is an AI-based application focused on design generation and user interaction, evaluation was primarily carried out through functional testing and user-based validation rather than traditional unit testing.

8.2.1 Testing Methodology

The system was tested using the following approaches:

- **Functional Testing:** Each module of the system was tested individually to ensure correct operation, including design generation, cost estimation, chatbot interaction, and PDF generation.
- **User Testing:** The system was evaluated by a group of users including students and general users to analyze usability, response quality, and overall satisfaction.
- **Performance Observation:** Response time, consistency of outputs, and system stability were monitored during repeated usage.

8.2.2 Test Participants

A total of **60 users** participated in the testing process. The participants included undergraduate students, developers, and non-technical users to ensure diverse feedback. The distribution of participants is shown in Table 8.1.

Category	Number of Users	Purpose
Students	30	Usability and feature testing
Developers	15	Functional validation
General Users	15	User experience evaluation
Total	**60**	

Table 8.1: User Distribution for Testing

8.2.3 Test Cases and Results

The following table 8.2 summarizes key functionalities tested and their evaluation ratings:

Test ID	Functionality	Rating (out of 5)	Remarks
TC1	Design generation from prompt	4.5	Generated consistent and realistic outputs
TC2	Multi-view generation	4.3	Views maintained structural consistency
TC3	Cost estimation	4.2	Accurate breakdown based on inputs
TC4	Budget optimization	4.1	Provided meaningful suggestions
TC5	Chatbot interaction	3.9	Context-aware responses observed but occasional delays
TC6	PDF report generation	4.4	Reports generated without errors
TC7	Sketch input handling	3.8	Worked well but sometimes required prompt refinement

Table 8.2: System Testing Results with Ratings

8.2.4 Performance Evaluation

The system performance was evaluated based on response time and output quality.

- Average design generation time: **5–8 seconds**
- Chatbot response time: **2–4 seconds**
- System uptime during testing: **Stable with no major failures**

8.2.5 Quantitative Performance Metrics

The overall system performance metrics are summarized in Table 8.3, which provides a quantitative evaluation based on user testing and observed system behavior.

Metric ID	Metric	Value	Description
PM1	Design generation success rate	92%	Percentage of successful image generations without failure
PM2	Multi-view consistency accuracy	89%	Consistency of structural elements across different views
PM3	Cost estimation reliability	85%	Accuracy of cost predictions based on input parameters
PM4	Chatbot response relevance	83%	Degree of meaningful and context-aware responses
PM5	System response efficiency	88%	Overall responsiveness of system under normal usage

Table 8.3: Quantitative Performance Metrics

8.2.6 User Feedback Analysis

Feedback collected from users indicates that:

- **85%** of users found the system easy to use.
- **82%** of users were satisfied with generated designs.
- **80%** of users found cost estimation helpful for planning.
- **78%** of users appreciated the chatbot assistance.
- **15%** of users reported minor inconsistencies in multi-view outputs.
- **12%** of users experienced slight delays in chatbot responses.
- **10%** of users suggested improvements in sketch-to-design accuracy.

8.2.7 Discussion

The testing results demonstrate that the GenArch AI system performs reliably across all major functionalities. The AI-generated outputs were consistent and visually meaningful, and the cost estimation module provided practical insights for users.

User feedback indicates that the system is intuitive and effective for both technical and non-technical users. However, minor limitations such as occasional inconsistencies in multi-view outputs and slight delays in chatbot responses were observed.

Although the system does not include traditional unit testing frameworks, the combination of functional validation, quantitative metrics, and user-based evaluation ensures that the application meets its intended objectives.

Overall, the testing process confirms that the system is stable, usable, and capable of delivering intelligent architectural design assistance with high reliability.

8.3 Challenges Faced

During development, several technical challenges were encountered:

- **Port Conflicts:** Default server ports were sometimes occupied by system services, requiring the backend server to run on an alternative port.
- **Multi-View Consistency:** Initially, different views produced inconsistent designs. This issue was solved by caching structural prompts and reusing them for additional views.
- **API Rate Limits:** Some AI services have request limits in the free tier. The system handled this by using fallback models and caching mechanisms.
- **Prompt Adaptation:** Vision models sometimes generated prompts specific to one view. A prompt adaptation method was used to adjust prompts for different camera perspectives.
- **Database Query Limitations:** Certain database queries required composite indexes. In-memory sorting was implemented as a fallback solution.
- **Image Size Optimization:** AI-generated images can be large, so compression techniques were applied to reduce storage and bandwidth usage.

8.4 Future Scope

The GenArch AI system can be further improved by introducing additional features.

- 3D model generation for interactive architectural visualization.
- Augmented reality (AR) support to place generated buildings in real environments.
- AI-based Vastu Shastra analysis for culturally aligned architectural planning.
- Automatic generation of 2D floor plans with room layouts.
- Integration with material suppliers for real-time cost estimation.
- Structural feasibility analysis using engineering AI models.
- Multi-language interface support for wider accessibility.
- Offline functionality using local caching techniques.

Chapter 9

Sustainability, Ethics and Societal Impact

This chapter evaluates the GenArchAI project from environmental, social, economic, and ethical perspectives in accordance with the updated NBA accreditation guidelines and GAPC 4.0 requirements. Modern engineering projects are expected not only to function correctly but also to consider sustainability, societal benefit, professional ethics, and long-term environmental impact. The GenArchAI system has been analyzed using Life Cycle Assessment, socio-economic evaluation, ethical compliance, and sustainable design principles to ensure responsible engineering practice.

9.1 Life Cycle Assessment (LCA)

Life Cycle Assessment (LCA) is used to evaluate the environmental impact of a system from its development stage to its usage and end-of-life disposal. Since GenArchAI is a software-based system, the environmental impact is mainly related to energy consumption, cloud infrastructure usage, and electronic device utilization rather than physical manufacturing.

9.1.1 Development Phase

The GenArchAI system is developed using software technologies such as React.js, Node.js, and cloud-based AI APIs. Unlike hardware projects, the development process does not require raw material extraction, industrial fabrication, or physical assembly. Therefore, the environmental impact during development is significantly lower.

However, development activities still consume electrical energy due to continuous use of computers, servers, and internet resources. The use of efficient programming practices, modular architecture, and optimized API calls helps reduce unnecessary processing and energy consumption.

9.1.2 Energy Consumption

The system uses cloud-based Artificial Intelligence models for image generation, vision analysis, and chatbot responses. These AI models run on remote data centers that require electrical power for computation and cooling. To reduce the environmental footprint, the following measures were considered:

- Image compression using the Sharp library to reduce file size
- Limiting API calls through prompt caching
- Using lightweight models when possible
- Avoiding unnecessary regeneration of images
- Rate limiting to prevent excessive server load

These optimizations help reduce the overall computational energy usage.

9.1.3 Deployment and Usage Phase

The application is deployed on cloud servers and accessed through web browsers. This approach avoids the need for dedicated hardware installation for each user. Multiple users can access the system using existing computers or mobile devices, reducing electronic waste.

Cloud deployment also allows efficient resource sharing, where server power is used only when required. This makes the system more energy efficient compared to traditional desktop software.

9.1.4 End-of-Life Considerations

Since GenArchAI is a software-based system, it does not produce physical waste at the end of its life cycle. Updates & improvements can be deployed digitally without replacing hardware. Data stored in cloud services can be removed safely without environmental damage.

This makes the project environmentally sustainable compared to traditional engineering products.

9.1.5 Environmental Impact Summary

- No physical manufacturing required
- Reduced paper usage due to digital design generation
- Efficient image compression reduces bandwidth consumption
- Cloud-based deployment minimizes hardware waste
- Optimized API usage reduces energy consumption

9.2 Socio-Economic Impact Analysis

Sustainability is not limited to environmental impact. It also includes the effect of technology on society, economy, and accessibility. The GenArchAI system provides significant benefits by making architectural design more affordable and accessible.

9.2.1 Cost Benefit Analysis

Traditional architectural design requires expensive software tools and professional services. Software such as AutoCAD, Revit, and SketchUp require costly licenses and long training periods. Hiring professional architects also increases project cost.

GenArchAI reduces these costs by providing AI-based design generation, cost estimation, and renovation visualization in a single platform. Users can generate designs instantly without purchasing expensive software. This reduces financial burden, especially for small-scale builders and homeowners.

9.2.2 Accessibility for Users

The system is designed for both technical and non-technical users. The interface uses simple controls, text prompts, and visual outputs. Users do not need professional knowledge of architecture to generate designs.

Since the system is web-based, it can be accessed from any device with internet connectivity. This makes the system usable in rural and urban areas without requiring specialized hardware.

9.2.3 Support for Small-Scale Developers

Small contractors and individual builders often do not have access to professional design tools. This system provides them with an affordable alternative for planning buildings, estimating cost, and visualizing designs.

Better planning reduces construction mistakes, saving time and money.

9.2.4 Educational Value

The project has strong educational benefits. Students can use the system to understand architectural concepts, AI technology, and cost estimation. Developers can learn modern web development, machine learning integration, and cloud computing.

9.2.5 Employment and Skill Development

The project encourages learning in modern technologies such as Artificial Intelligence, Web Development, Cloud Computing, and Data Processing. These skills are important for future engineering careers.

The project also shows how software systems can be used to solve real-world problems.

9.3 Ethical Considerations and Compliance

Professional ethics are essential in engineering practice. The GenArchAI system is designed with responsibility, transparency, and security in mind.

9.3.1 Data Privacy

The system does not store personal user information except project data required for operation. API keys are stored securely in server environment variables and are never exposed to the frontend. This prevents misuse of external services.

9.3.2 Responsible Use of Artificial Intelligence

AI-generated images are provided only for visualization purposes. The system clearly states that generated designs should not be used directly for construction without verification by professional engineers or architects.

This prevents misuse of the system and ensures user safety.

9.3.3 API Usage Compliance

All external services used in the project follow their official terms of service. The APIs used include:

- Google Gemini API
- Hugging Face API
- Groq API
- Firebase Firestore
- Cloudinary Storage

These services allow academic and development use under their free-tier licenses.

9.3.4 Security Measures

To ensure safe operation, the system includes the following security features:

- API keys stored in environment variables
- Rate limiting to prevent misuse
- File size validation for uploads
- Error handling to avoid server crashes
- Input validation for user data

9.3.5 Professional Integrity

All libraries, frameworks, and APIs used in the project are properly acknowledged. The project follows academic honesty and avoids plagiarism. The implementation is original and developed for educational purposes.

9.4 Sustainable Design Optimization

During development, several design decisions were made to reduce resource usage and improve efficiency.

9.4.1 Efficient Image Processing

AI-generated images can be large in size. The system compresses images using the Sharp library before storing them. This reduces storage space, bandwidth usage, and server load.

9.4.2 Cloud-Based Architecture

Using cloud services avoids the need for dedicated hardware. This reduces energy consumption and electronic waste. Cloud servers also allow efficient sharing of resources.

9.4.3 Reusable Components

The frontend is designed using reusable components. This reduces code duplication and improves maintainability. Reusable design also reduces development time and resource usage.

9.4.4 Design for Scalability

The system is modular and divided into routes, services, and utility modules. This allows new features to be added without rewriting the entire program. Modular design reduces future development effort.

9.4.5 Design for Disassembly (Software Perspective)

In hardware systems, design for disassembly allows parts to be replaced easily. In software, this is achieved by modular programming. Each module in GenArchAI can be modified independently without affecting the whole system. This improves long-term sustainability.

9.5 Standards and Guidelines Followed

The project follows standard software engineering and ethical guidelines.

- IEEE Software Engineering Standards
- REST API design principles
- Secure API key storage practices
- Cloud deployment best practices
- Modular programming principles
- NBA Outcome Based Education guidelines
- GAPC 4.0 sustainability requirements

9.6 Conclusion

The GenArchAI project demonstrates that modern software systems can be developed with consideration for environmental sustainability, social responsibility, and ethical engineering practice. The system reduces the need for physical resources, improves accessibility to architectural tools, and follows secure and responsible development methods.

By applying Life Cycle Assessment, socio-economic analysis, ethical compliance, and sustainable design principles, the project satisfies the NBA Programme Outcomes related to environment, society, and professional ethics. This shows that the system is not only technically successful but also socially responsible and environmentally sustainable.

Chapter 10

Conclusion and Future Work

10.1 Conclusion

GenArchAI demonstrates the successful integration of multiple AI technologies into a single architectural design platform. The system combines image generation, vision analysis, and conversational AI to enable users to create and visualize building designs efficiently.

The platform supports multi-view architectural generation using prompt caching to ensure consistency across different perspectives. It also includes a cost estimation module tailored to the Indian construction market, making it practical for real-world applications. The system architecture is designed to be fault-tolerant with fallback mechanisms, ensuring reliability during API limitations.

Additional features such as PDF report generation and real-time collaboration using Socket.IO enhance usability and user experience. Overall, GenArchAI makes architectural visualization more accessible to students, homeowners, and designers by reducing dependency on expensive software and professional expertise.

Furthermore, the project highlights the potential of integrating Generative AI into real-world applications beyond traditional domains. By combining multiple AI services within a unified workflow, the system demonstrates how complex design tasks can be simplified into an interactive and user-friendly process.

In addition, GenArchAI serves as a scalable and extensible platform that can be enhanced with future technologies. Its modular architecture allows new features to be integrated easily, making it a strong foundation for next-generation intelligent architectural systems.

10.2 Future Work

Although GenArchAI provides a comprehensive solution for architectural design and visualization, there are several areas where the system can be further improved. Future enhancements can include the implementation of a secure user authentication system and design version history, allowing users to manage projects more effectively and track design changes over time.

The platform can also be extended to support advanced features such as 3D model generation and automatic floor plan creation, which would significantly improve spatial understanding and practical usability. Integration of Vastu compliance checking can make the system more suitable for Indian users by aligning designs with cultural and traditional guidelines.

Additionally, technologies such as Augmented Reality (AR) and Virtual Reality (VR) can be incorporated to provide immersive visualization experiences. Structural validation using engineering principles and support for CAD export formats would enhance the system's applicability in real construction workflows. Integration with material marketplaces can further improve cost estimation accuracy by providing real-time pricing data.

Overall, these future improvements will transform GenArchAI into a more powerful, intelligent, and industry-ready architectural design platform.

References

[1] M. Shi, J. Seo, S. H. Cha, B. Xiao, and H.-L. Chi, *"Generative AI-powered architectural exterior conceptual design based on the design intent,"* J. Comput. Des. Eng., vol. 11, pp. 125–142, 2024, doi: 10.1093/jcde/qwae077.

[2] J. Ko, J. Ajibefun, and W. Yan, *"Generative AI-powered parametric modeling and BIM for architectural design and visualization,"* Proc. of the Design Society, vol. 5, ICED25, pp. 1943-1952, Aug. 2025, doi: 10.1017/pds.2025.10208.

[3] J. A. Ujong, E. M. Mbadike, and G. U. Alaneme, *"Prediction of cost and duration of building construction using artificial neural network,"* Asian Journal of Civil Engineering, vol. 23, no. 7, pp. 1117–1139, Jul. 2022, doi: 10.1007/s42107-022-00474-4.

[4] P. Li, B. Li, and Z. Li, *"Sketch-to-Architecture: Generative AI-aided Architectural Design,"* arXiv preprint arXiv:2403.20186 [cs.CV], Mar. 2024.

[5] K. Yim, *"AI Chatbots in Architectural Design: Exploring the Potential and Limitations in Conceptual Image Development,"* International Journal for Infonomics (IJI), vol. 17, no. 1, pp. 2207-2214, 2024, doi: 10.20533/iji.1742.4712.2024.0232.

[6] J. Chen, D. Wang, Z. Shao, X. Zhang, M. Ruan, H. Li, & J. Li (2023). *Using artificial intelligence to generate master-quality architectural designs from text descriptions.* Buildings, 13, 2285, doi: 10.3390/buildings13092285.

[7] P. Anzalone, & C. Clarke (2003). *Architectural applications of complex adaptive systems.* In *Proceedings of the 2003 Annual Conference of the Association for Computer Aided Design in Architecture.* Ball State University.

[8] V. Baltus, & T. Žebrauskas (2019). *Parametric design concept in architectural studies.* Architecture and Urban Planning, 15, 96–100, doi: 10.2478/aup-2019-0013.